AQA GCSE Geography (8035) — Paper 2: Challenges in the Human Environment · 3.2.3 The Challenge of Resource Management

2.4 Resource Management

1. The Significance of Food, Water and Energy

Food, water and energy are fundamental resources. Everyone needs them for **economic and social wellbeing**:

- **Food** — a healthy, balanced diet is needed for good health, growth and the ability to work and learn. Undernourishment reduces productivity and leaves people vulnerable to disease.
- **Water** — needed for drinking, sanitation, food production and industry. Unsafe water spreads diseases such as cholera and typhoid.
- **Energy** — needed for heating, cooking, lighting, transport and industry. Access to electricity allows businesses to grow and children to study after dark.

Resource security — having reliable access to enough affordable, good-quality resources to meet needs. **Resource insecurity** is when this access is not guaranteed.

2. Global Inequalities in Supply and Consumption

Resource	Global pattern
Food	Enough food is produced to feed everyone, but around 700 million people are undernourished, mostly in sub-Saharan Africa and South Asia. People in HICs consume more than 3,400 calories a day on average; in some LICs it is under 2,200. About a third of food produced is wasted.
Water	About 2 billion people lack safely managed drinking water. HICs use far more per person: an American uses around 300–400 litres a day, a UK resident about 140 litres, but many

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	people in sub-Saharan Africa have less than 20 litres.
Energy	Around 700 million people (mostly in sub-Saharan Africa) have no electricity. The richest countries consume many times more energy per person than the poorest; China, the USA and India are the largest consumers overall.

Why the inequality? Wealth (ability to pay), climate and physical geography (rainfall, fertile soil, fossil fuel reserves), technology, conflict and political stability.

3. Food in the UK

Growing demand for exotic foods and all-year produce

- UK consumers now expect foods such as avocados, mangoes, coffee and out-of-season strawberries all year round. The UK imports around **40–45% of its food**.
- Reasons: rising incomes, foreign travel, multicultural population, supermarket competition and cheap air freight.
- Imports can benefit producers in LICs (e.g. Kenyan green beans and flowers) but use up water and land needed locally.

Food miles and carbon footprints

> **Food miles** — the distance food travels from where it is produced to where it is consumed.
> **Carbon footprint** — the total greenhouse gas emissions caused by producing, transporting and selling a product.

- Air-freighted food (e.g. Peruvian asparagus) has a very high carbon footprint. However, food miles do not tell the whole story — tomatoes grown in heated UK greenhouses can produce more emissions than tomatoes grown in the Spanish sun and trucked to the UK.

Moving towards local sourcing and organic food

- Farmers' markets, farm shops, veg-box schemes and "grow your own" allotments reduce food miles and support local farmers.

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- **Organic** produce (grown without artificial fertilisers or pesticides) has grown in popularity, but it is more expensive and yields are lower.

Agribusiness

> **Agribusiness** — intensive farming run as a large commercial business, using machinery, chemicals and technology to maximise output and profit.

- Farms in East Anglia, for example, are large, highly mechanised and specialised in cereals and vegetables, often under contract to supermarkets.
- **Benefits:** cheap, plentiful food; efficient production. **Problems:** loss of hedgerows and wildlife, soil degradation, pollution from fertiliser run-off, fewer farm jobs.

4. Water in the UK

Changing demand

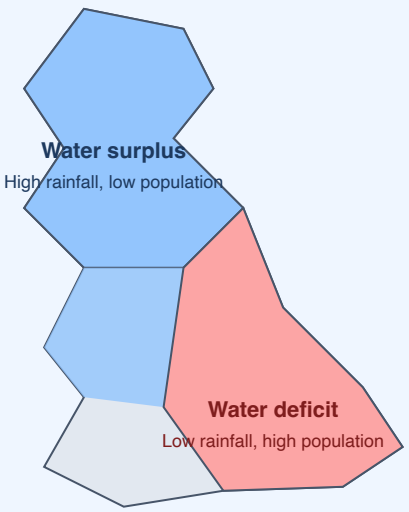
- UK water demand is expected to rise, driven by population growth (especially in the South East), more households (smaller households use more water per person), and appliances such as dishwashers and power showers.
- Climate change is expected to bring hotter, drier summers, reducing supply when demand is highest.

Water quality and pollution management

- Sources of pollution: fertiliser and pesticide run-off from farms, sewage overflows (storm overflows discharged untreated sewage into rivers and seas hundreds of thousands of times a year in recent years), industrial discharges and plastic waste.
- Management: the Environment Agency monitors water quality; water companies treat water; stricter laws limit farm run-off (Nitrate Vulnerable Zones); the Thames Tideway Tunnel ("super sewer", completed 2025) captures sewage overflows in London; fines for companies that pollute.

Matching supply and demand — areas of deficit and surplus

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Simplified map of UK water surplus and deficit

- The **north and west** (Scotland, Wales, northern England) have high rainfall (over 1,500 mm a year in uplands) and lower populations — a **water surplus**.
- The **south and east** (London, the South East, East Anglia) have low rainfall (around 600 mm) and high populations — a **water deficit**. Parts of the South East have less water available per person than some Mediterranean countries.

Water transfer to maintain supplies

- **Water transfer** moves water from areas of surplus to areas of deficit, via pipelines, canals or rivers. Examples: the Elan Valley reservoirs in Wales supply Birmingham through a 118 km aqueduct (since 1904); Kielder Water in Northumberland can transfer water to the Tees, Wear and Tyne.
- A national water grid has been proposed but is very expensive. Issues include the cost of pumping, environmental impacts of mixing water from different river basins, and political opposition (e.g. in Wales).

5. Energy in the UK

The changing energy mix

Source	Change
Coal	Supplied around 40% of electricity as recently as 2012; the last coal power station (Ratcliffe-on-Soar) closed in September 2024.

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Natural gas	Now the largest single source of electricity (roughly a third) and the main fuel for home heating.
Renewables	Grew from under 5% of electricity in 2005 to around half today — wind (especially offshore) is now the largest renewable source; also solar, biomass and hydro.
Nuclear	Around 13–15% of electricity; several older plants are closing. Hinkley Point C (Somerset) is under construction and Sizewell C has been approved.

Reduced domestic supply of fossil fuels

- North Sea oil and gas production peaked around 1999–2000 and has declined since. The UK now imports around half of its gas (by pipeline from Norway and as liquefied natural gas by ship), making it vulnerable to world price rises, as in 2022.

Economic and environmental issues of exploiting energy sources

Source	Economic issues	Environmental issues
Fracking (shale gas)	Could reduce imports and create jobs	Risk of earth tremors (Preston New Road, Lancashire, 2019) and groundwater pollution; a moratorium has been in place in England since 2019
Nuclear	Very expensive to build and decommission (Hinkley Point C is expected to cost over £30 billion); reliable low-carbon power	Low carbon emissions but radioactive waste must be stored safely for thousands of years
Wind	Costs have fallen dramatically; creates jobs (e.g. turbine factories in Hull); intermittent supply	Low carbon; visual and noise impact onshore; risk to birds; offshore farms such as Hornsea (the world's largest) avoid many land conflicts
North Sea oil and gas	Jobs (especially around Aberdeen) and tax revenue; declining reserves are more costly to extract	Greenhouse gas emissions; risk of oil spills

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Exam Tip: Paper 2 asks about the UK for all three resources, then you answer in more depth on **one** of food (2.5), water (2.6) or energy (2.7). Make sure you revise this overview topic as well as your chosen option.

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